LESSON PLAN · SCRATCH (GR 6–8) BEGINNER: CODE FOUNDATIONS

Lesson 6: Project: Generative Art

Module 2 | 50 minutes | Project: nested loops and randomness | Grades 6–8 · ages 11–14

Objective: Plan, build and test a generative-art program that uses nested loops, variables and randomness to draw an original design.

Success criteria

- My art uses nested loops.
- It uses randomness in a controlled way.
- I can explain my rules in an artist statement.

Key vocabulary

- **nested loop**: A loop inside another loop.
- **generative art**: Art made by following rules in code.
- **randomness**: Values chosen by chance.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)
- Printed project brief

Timed lesson plan (50 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Rule art (5 min): roll a die to choose shape size and draw a rosette by hand.
5 min	**Watch & plan**	Play the project video; learners plan on the project brief.
25 min	**Build**	Build-along steps, then Plan the rules. Check in halfway: everyone has something working.
7 min	**Test & peer feedback**	Swap and test each other's work: two stars and a wish.
8 min	**Improve & submit**	Improve one thing; Project: Generative Art.

Unplugged warm-up

Rule art (5 min): roll a die to choose shape size and draw a rosette by hand.

Teaching the idea

Nested loops compose shapes into patterns; controlled randomness creates variety within rules. Artist statements build the habit of explaining algorithms.

Analogy: A kaleidoscope: simple rules and a little chance make endless patterns.

Live demo: Build a rosette live; then add random positions and run it three times.

Questions to ask

Ask	Listen for
Which loop draws the shape?	The inner loop.
What does the outer loop do?	Repeats and turns the shape.
What did you keep fixed?	Artistic choice.

Differentiation

Support
Start from the polygon machine code.

Stretch
Change the pen colour inside the outer loop for a rainbow rosette.

Common misconceptions

Misconception	How to fix it
Random means messy	Fixed rules keep the design coherent.
Nested loops run once each	The inner loop runs fully for every outer repeat.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Loops	No loops	Single loop	Nested loops	Nested loops with variables
Randomness	None	Random but chaotic	Controlled randomness	Thoughtful mix of random and fixed
Visual design	Unclear	Simple	Attractive	Striking and original
Code quality	Hard to follow	Works	Tidy and DRY	Elegant and easy to change
Artist statement	Missing	Brief	Explains the rules	Explains rules and design choices

Exit ticket
How many times does the inner block run in repeat 12 { repeat 6 { … } }?

Homework
Run your art five times and pick your favourite version. Why that one?

Answer key — Kitabash activities

1. Plan the rules (perform · ordering)

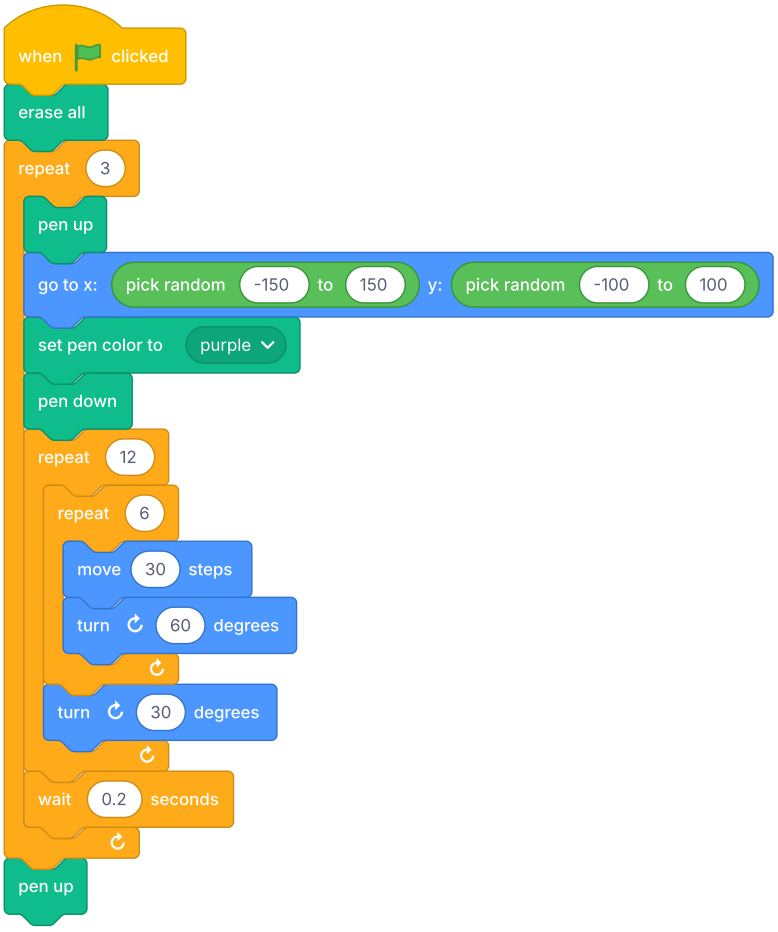
Order: Erase all and set the pen → Choose a random position → Inner loop draws one shape → Outer loop turns and repeats the shape → Repeat for more rosettes

2. Project: Generative Art (probe · code lab)

Goals checked: Nested loops · Randomness · A rich design · Starts clean

Example project:

Pen



Answer key — worksheet

1. **Key words:** nested loop = A loop inside another loop. · generative art = Art made by following rules in code. · randomness = Values chosen by chance.
2. **Plan the rules:** 1 Erase all and set the pen · 2 Choose a random position · 3 Inner loop draws one shape · 4 Outer loop turns and repeats the shape · 5 Repeat for more rosettes
3. **My plan:** A clear plan that covers every requirement.
4. **Checklist:** All requirements met.
5. **Two stars and a wish:** Kind, specific feedback and an improvement.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.